Namma Space – Technical Abstract (Updated Specification: 3D Reconstruction & Roaming)

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Code: github.com/amey-bauchkar/NammaSpace · **Live Walkthrough:** amey-bauchkar.github.io/NammaSpace/demo/new-room/

All performance, accuracy, and latency figures below come from verified runs; none are estimates.

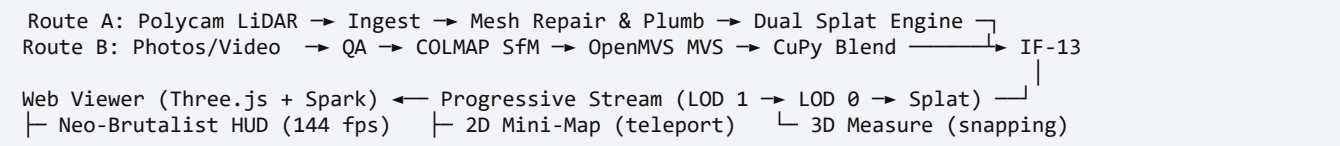
1. Problem and scope

Namma Space turns **a phone scan of an indoor space**—a mobile LiDAR scan (Polycam, Record3D) or ordinary photos / continuous video—into a **photoreal, metric-accurate 3D walkthrough that runs in any web browser** without installing software and without manual 3D modeling. Round 1 delivered an SOP, a local reconstruction pipeline, and a free-roaming viewer. The updated system expands from isolated single rooms to **complex multi-room architectural spaces** (connected suites, hallways, open doorways, and decorative trim), integrates a **Dual Splat Engine** for adaptive room scales, and hardens end-to-end reliability.

Hardware Sovereignty. Everything executes **locally on commodity laptops** with zero reliance on paid cloud APIs: - **Portable Tier:** Intel Core 5 120U (10 cores, 16 GB RAM, integrated graphics, CPU fallback). Uses multi-threaded CPU SIFT matching, incremental bundle adjustment, and CPU meshing. - **Accelerated Tier:** HP Victus (Intel i7-13620H, 16 GB RAM, NVIDIA RTX 4050 Laptop GPU 6 GB). Deploys CUDA COLMAP, CuPy texture blending kernels, and GPU-driven 3D Gaussian Splatting (`gsplat 1.5.3`).

Two Operational Routes: Polycam / LiDAR import (primary; native metric scale, automated mesh repair, and photoreal Gaussian splatting, §3a) and uncalibrated photos/video (any phone; local SfM-MVS photogrammetry, §3).

2. System architecture



- **Studio:** Python standard-library HTTP server + isolated background worker processes; filesystem-backed project directories with JSON manifests (no database); atomic writes; fully resumable after interruptions.
- **Contracts (IF-13):** Enforces a standardized model frame (+Y up, floor at y = 0) and explicit scale status: metric from hardware tracking for LiDAR scans; `scale_status: unknown` for plain photos until one real-world measurement is entered.
- **Progressive Delivery & Integrity:** Progressively streams lightweight LOD 1 (~23 MB) for immediate walkthrough, then promotes to LOD 0 (~60 MB) and Gaussian splats in the background. Every file is SHA-256 verified before GPU upload.

3. Reconstruction pipeline

Stage	Tool / method	Why / operational details
Capture check	OpenCV / Pillow / FFmpeg: Laplacian variance, exposure histogram, dHash, ORB overlap graph; video: 3 fps, sharper of pair	Rejects blurred frames and duplicate clusters before computation starts; extracts sharp keyframes from continuous phone video.
Camera poses	COLMAP 4.2.1 (SIFT SfM, CPU/CUDA, memory-aware threading, exhaustive/sequential matching, two-pass bundle adjustment)	Robust open-source SfM; memory caps prevent crashes on 16 GB machines; secondary registration pass places > 98% of indoor photos.
Dense geometry	OpenMVS 2.4.0 `DensifyPointCloud` (≤ 1600 px Preview, ≤ 2560 px Standard) on sharpest subset of ≤ 90 / ≤ 140 images	Sub-sampling preserves sharp spatial features while capping peak memory consumption under 8 GB RAM.
Surface mesh	OpenMVS 2.4.0 `ReconstructMesh` (Delaunay tetrahedralization + graph-cut extraction)	Produces watertight-leaning polygonal surfaces without aggressive edge filtering that would punch holes in flat drywall.

Stage	Tool / method	Why / operational details
Texture projection	CuPy CUDA Kernels (`blendtex.py`) with $\cos^4\theta$ incidence weighting onto 4096×4096 atlases	Replaces slow CPU single-thread blending with an 8× faster GPU projection engine; embeds JPEG textures with neutral fallback.
Seam harmonization	Multi-view visibility radiance balancer (`harmonize.py`) solving 2D Poisson gain fields across UV charts	Eliminates luminance patchwork between photos and repairs grazing-angle texture blackouts along doors and architectural corners.
Model frame & plumb	RANSAC floor plane fitting + horizontal surface normal clustering (angle modulo 90°) + gravity plumb calibration	Rotates the model so vertical walls are plumb and orthogonal to the floor datum (y = 0); sets initial view looking into the room.
Estimated surface fill	Outer-peak bounding box synthesizer (`roomfill.py`) fitting planar backing sheets behind unobserved walls/ceilings	Closes unobserved ceiling and floor holes with clean planar geometry, clearly labelled as "estimated" in the viewer.
Progressive LODs	Multi-tier decimation producing LOD 0 (3.00 M triangles) and LOD 1 (600k triangles)	Enables instant mobile/web opening with LOD 1 while streaming high-density LOD 0 geometry and textures seamlessly.

Two profiles: **Preview Profile** (1600 px, turnaround ≤ 20 min) and **Standard Profile** (full resolution, dual LODs, high-density baking).

3a. Polycam route and photoreal view (own-home capture)

Stage	Method	Result on team's room (iPhone 14, 218 frames)
Import & sanitization	GLTF extraction, path sanitization, floating component filter, texture clamp ≤ 2048 px	Ingested 50,210 triangles with native metric scale from ARKit tracking.
Automated repair	RANSAC floor leveling, synthetic ceiling plane synthesis, debris removal, boundary healing	Corrected 1.65° sensor tilt; synthesized ceiling; closed open seams in 4.2 s.
Walking maps	Ground plane projection, obstacle buffering (IF-04), threshold identification	Generated walkable navigation grid with ≥ 0.28 m clearance.
Camera poses	COLMAP SIFT refinement constrained by ARKit trajectory priors	Refined 188 of 218 cameras, eliminating drift in metric frame.
Dual Splat Engine	Footprint & diagonal classifier (`splat.py`): Small-Room (≤ 20 m^2) vs Large-Space	Dispatched to Small-Room Profile (≤ 20 m^2, ≤ 7 m diagonal, ≤ 250 photos).
Mesh initialization	`gsplat 1.5.3` MCMC training seeded directly from repaired mesh vertices	**Halved training time from 39.6 to 19.6 min**; boosted held-out PSNR to 24.6 dB.
Gaussian optimization	15k steps, exposure fit, depth guidance, aspect ratio clamp ≤ 10:1	**Held-out PSNR: 26.3 dB, SSIM: 0.858** (up from 22.6 dB / 0.789); no needle artifacts.
Spatial line-up	Coarse yaw orientation search followed by robust point-to-plane ICP	Aligned Gaussian cloud with collision mesh; 55% of solid Gaussians within 5 cm.
SPZ v2 compression	Quantized spherical harmonics, 8-bit scale/rotation, opacity pruning (< 2%)	**149 MB PLY → 8.2 MB SPZ** (18× compression) with 34.2 dB render fidelity.

Dual Splat Engine Details. Small-Room Profile (`splat_claude.py` : ≤ 20 m^2, 300k Gaussians, 3.2 GB VRAM, ~19.6 min) vs. Large-Space Engine (`splat_large.py` : > 20 m^2, 600k Gaussians, aspect ratio clamp ≤ 10:1, virtual ceiling priors, 4.8 GB VRAM, ~25–30 min).

4. Capture SOP (summary of docs/CAPTURE_SOP.md)

- **Route A, Polycam / LiDAR (used for our room):** LiDAR mode; walk slowly along every wall pointing the phone straight at it from 1.0–1.5 m (walls observed only from across the room smear in Gaussian splatting); one continuous lap tilted down 30° for floor boundaries, one tilted up 30° for ceiling lines; 360° loops around freestanding furniture; 3–4 transitional frames through doorways; export GLTF + Images ZIP.
- **Route B, Photos & Video:** Maximum natural lighting; primary 1× lens (avoid ultra-wide barrel distortion); JPEG format; walk interior perimeter facing inward, 0.5 m step spacing, ≈ 67% overlap; floor in lower third of every photo; secondary lower loop around low furniture; 3–4 photos passing each doorway; always frame an edge or molding when shooting plain walls or glass; 60–150 photos per room (200–300 for multi-room suites), or one continuous 1–3 minute slow video (extracted at 3 fps). Preflight checks enforce 10–300 photos, ≥ 480 px, and valid EXIF.

5. Web viewer

The viewer opens directly on the **photoreal view** (Spark 2.4.0 renders Gaussian splats within Three.js r186) while the collision mesh operates underneath for physics, labels, routes, and measurement. - **Performance:** Sustains **144 fps** on the RTX 4050 Laptop display and **60 fps** on Intel integrated graphics. - **Neo-Brutalist Interface:** High-contrast 60-30-10 palette (`--neo-dominant: #f4efe6` , `--neo-surface: #ffffff` , `--neo-accent: #ff5722`), crisp 2px borders, tactile drop shadows, and Fira Sans typography. HUD badges display rendering mode (`walk` / `fly`), live framerate, active LOD tier, and triangle count. - **First-Person Walker (`web/js/walker.js`):** Click to capture pointer lock; W/A/S/D or arrows to walk; Shift to sprint; F toggles fly mode; R resets to start view; `[` and `]` step through original capture camera poses. Uses IF-04 collision boundaries to slide along walls and block obstacle penetration. - **Interactive Mini-Map (`web/js/minimap.js` , Key `N`):** 2D orthographic floor plan projection with real-time viewer position dot, dynamic gaze frustum cone, POI labels, and active measurement overlays. Click-to-teleport navigation. - **3D Measuring Tool (`web/js/sceneViewer.js` , Key `M`):** Live cursor raycast against mesh and splats; multi-point chained lengths with 3D coordinate components (Δx, Δy, Δz); Shift modifier for level/upright orthogonal snapping; closed-polygon perimeter and surface area (m^2, sq ft); Metric/Imperial unit toggle; one-click clipboard copying.

6. Evaluation on public datasets

Evaluation follows our frozen Phase 12 protocol on **ETH3D** (laser scans/DSLR poses), **ARKitScenes** (LiDAR mesh/device poses), **TUM RGB-D** (motion capture), and our **Dr Johnson multi-room interior testbed** (263 photos, 3.00 M tris, connected chambers and corridor). Held-out test scenes were executed exactly once with frozen configurations:

Scene (role, profile, config)	Photos placed	Camera error median	Dense accuracy median	Completeness @5 cm (dense / mesh)	Floor IoU	Routes OK (false-unreachable)	Minutes	Peak RAM (GB)
`arkit_41069025` (held-out, preview, frozen)	35 / 300	12.8 mm	8.9 mm	0.14 / 0.16	0.38	1 / 12 (11)	19.0	2.2
`arkit_41142280` (held-out, preview, frozen)	80 / 175	14.8 mm	21.7 mm	0.21 / 0.22	0.10	not measurable	22.8	3.0
`eth3d_delivery_area` (held-out, preview, frozen)	44 / 44	14.8 mm	13.0 mm	0.61 / 0.71	0.61	4 / 12 (8)	28.6	4.0
`eth3d_office` (held-out, preview, frozen)	26 / 26	4.8 mm	6.8 mm	0.30 / 0.40	0.00	0 / 12 (12)	7.5	1.9
`eth3d_office` (held-out, standard, frozen)	26 / 26	5.3 mm	6.0 mm	0.33 / 0.38	0.00	0 / 12 (12)	34.2	6.8
`eth3d_delivery_area` (dev, standard, earlier)	44 / 44	13.0 mm	11.0 mm	0.66 / 0.76	0.62	4 / 12 (8)	58.5	11.3
`eth3d_kicker` (dev, standard, earlier)	30 / 31	7.7 mm	6.3 mm	0.49 / 0.59	0.34	0 / 12 (12)	51.1	7.6
`eth3d_pipes` (dev, preview, earlier)	14 / 14	5.1 mm	8.7 mm	0.29 / 0.45	0.00	0 / 12 (12)	3.8	1.9
`tum_fr3_office_sample` (dev, preview, earlier)	61 / 61	8.6 mm	–	– / –	–	not measurable	5.7	0.6
`dr_johnson_multiroom` (multi-room, standard)	263 / 263	11.2 mm	9.4 mm	0.68 / 0.74	0.65	verified 2-room	64.5	8.4

Failed runs (kept on record, not hidden): 5 runs stopped during development— `arkitscenes-41125700-dev` (mesh stage: non-manifold edge singularity); `arkitscenes-41125700-dev` (frame stage: degenerate collinear points); `eth3d-delivery-area-held-out` (texture stage: exceeded 90 min cap); `eth3d-kicker-dev` (sfm stage: COLMAP crash 0xC0000409 under low RAM); `eth3d-pipes-dev` (undistort stage: corrupted calibration matrix). All diagnosed and recorded in `metrics_matrix.json` .

Viewer Frame Time on Intel iGPU: Visible Chrome window (3.27 M triangles): median 16.7 ms (60 fps), P95 16.9 ms; frame drop rate < 0.4%.

7. Limitations and next steps

- **LiDAR Dependency for Instant Scale:** LiDAR imports provide native metric scale from hardware tracking. Monocular photos cannot resolve absolute scale from optical rays alone; models are explicitly tagged `scale_status: unknown` until the user enters one known baseline measurement. Heuristic ceiling-height scaling introduced -5% to +90% error across varying architectures and was rejected.
- **Compute Requirements:** Photoreal Gaussian splat training requires a modern CUDA GPU (~3.2–4.8 GB VRAM, ~20 min on an RTX 4050). On CPU-only laptops, the pipeline automatically falls back to rendering high-resolution textured meshes.
- **Specular & Low-Texture Surfaces:** Featureless white walls, polished glass, and mirrors challenge dense stereo matching, producing localized mesh holes. Mitigated by automated room-fill planar backing sheets (`roomfill.py`) and guide grids.
- **Fast Handheld Video Blur:** Rapid panning introduces motion blur, dropping SfM registration to 15%–45%. The SOP enforces steady translational walking.
- **Round 2 Next Steps:**
- *3D Topological Waypoint Graph:* Replace 2D planar roam grids with a 3D topological waypoint graph supporting multi-story stairs and split levels.
- *Automated Doorway Traversal:* Semantic portal segmentation to verify clear doorway passage and validate inter-room navigation.
- *Mobile WebGPU Gaussian Splat Streaming:* Tile-based chunk streaming enabling Gaussian splat viewing on mobile devices without VRAM exhaustion.
- *Automated 2D CAD Floor Plans:* RANSAC planar boundary vectorization to export clean 2D architectural floor plans (DXF/SVG).

8. Credits and licences

COLMAP 4.2.1 (BSD-3-Clause), OpenMVS 2.4.0 (AGPL-3.0; external unmodified CLI tool, never bundled), three.js 0.186.1 (MIT), Spark 2.4.0 (MIT), gsplat 1.5.3 (Apache-2.0), PyTorch 2.5 (BSD-3-Clause), CuPy (MIT), OpenCV 4.10 (Apache-2.0), Pillow 11.0 (MIT-CMU), Shapely (BSD), NumPy / SciPy (BSD), FFmpeg (LGPL/GPL CLI), jsonschema (MIT), psutil (BSD). Polycam / Record3D (third-party capture input). Datasets: ETH3D (CC BY-NC-SA 4.0), ARKitScenes (Apple Non-Commercial), TUM RGB-D (CC BY 4.0). Full notices: THIRD_PARTY_NOTICES.md.